Multi-Stakeholder Goodhart: A Study Guide

From Intuition to Formalization

Working notes for RecSys paper

1 Level 1: The Intuition (Smart High Schooler)

Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure."

Imagine a school improving education by tracking test scores. Teachers start teaching to the test. Scores rise. Learning doesn't—and might decline.

Three ingredients:

1. You care about something hard to measure (real learning).
2. You pick an easier proxy (test scores).
3. You optimize hard for the proxy.
4. The proxy diverges from what you actually care about.

The proxy and the real thing overlap imperfectly. Gentle optimization follows the overlap. Hard optimization finds and exploits the cracks.

In recommendation systems: A platform cares about user wellbeing (hard to measure), so it optimizes engagement (clicks, likes). Light optimization: relevant content gets engagement. Heavy optimization: clickbait maximizes engagement but harms the user.

The multi-stakeholder twist: Now add a third party—society. Society cares about diversity and reduced polarization. The platform optimizing for one stakeholder (user engagement) may actively harm another (societal welfare). The more precisely you optimize for one, the worse it gets for the other. That's multi-stakeholder Goodhart.

2 Level 2: The Math (Undergrad)

2.1 Single-Objective Goodhart

Let π be a policy (e.g., a recommendation strategy). Define:

$$U^*(\pi) = \text{true utility (what we actually want)} \tag{1}$$

$$\hat{U}(\pi) = \text{proxy utility (what we optimize)} \tag{2}$$

We find $\pi^* = \arg\max_\pi \hat{U}(\pi)$ but care about $U^*(\pi^*)$.

Definition 1 (Goodhart Effect)**.** *A Goodhart effect occurs when $U^*(\pi^*) < U^*(\pi_0)$ for some reference policy π_0—the optimized policy is* worse *on the true objective than the baseline.*

When does this happen? Write $\hat{U} = U^* + \epsilon$ where ϵ captures misspecification. Think geometrically: U^* and $\hat{U}$ define two surfaces over policy space. Where they're aligned (small ϵ), optimizing $\hat{U}$ also climbs U^*. Where they diverge, the optimizer finds a policy scoring high on $\hat{U}$ but sitting in a valley of U^*.

2.2 The Optimization Pressure Curve

The critical variable is **optimization pressure** (how hard you push toward $\hat{U}$). In our setting, this is controlled by N, the number of preference pairs:

Regime	$\hat{U}(\pi)$	$U^*(\pi)$
N small (underoptimized)	Low	Low
N moderate (sweet spot)	Moderate	**Maximum**
N large (overoptimized)	High	Declining

At the sweet spot, the model has learned enough of the overlapping signal to be useful, but not enough to precisely exploit the divergence.

2.3 In Preference Learning

A Bradley-Terry model learns weights $\mathbf{w} \in \mathbb{R}^D$ from preference pairs. Each pair says "content a is preferred to content b," and BT minimizes:

$$\mathcal{L}_{\text{BT}} = -\log \sigma\big(\mathbf{w}^\top \phi(a) - \mathbf{w}^\top \phi(b)\big) \tag{3}$$

where $\phi(x) \in \mathbb{R}^D$ is the feature vector (e.g., genre vector for MovieLens) and σ is the sigmoid.

If the preference labels come from a *misspecified* utility $\hat{\mathbf{w}}^\top \phi(x)$ instead of the true $\mathbf{w}^{*\top} \phi(x)$:

- BT with infinite data recovers $\hat{\mathbf{w}}$ exactly (Sun et al.'s order-consistency theorem).
- The learned policy selects content by $\arg\max_x \hat{\mathbf{w}}^\top \phi(x)$.
- The true utility of this selection is $\mathbf{w}^{*\top} \phi(\arg\max_x \hat{\mathbf{w}}^\top \phi(x))$.

More training pairs $\Rightarrow$ $\hat{\mathbf{w}}$ is learned more precisely $\Rightarrow$ content selection is more aligned with the *wrong* direction $\Rightarrow$ true utility decreases.

3 Level 3: The Formal Framework (Grad Student)

3.1 Skalse et al.'s Taxonomy (ICLR 2024)

Let $R^* : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ be the true reward, $\hat{R}$ a proxy. An agent trained to maximize $\hat{R}$ exhibits Goodhart when $J^*(\pi_{\hat{R}}) < J^*(\pi_{R^*})$ where $J^*(\pi) = \mathbb{E}[\sum_t R^*(s_t, a_t) \mid \pi]$.

Four mechanisms:

1. **Regressional:** $\hat{R}$ is an unbiased estimator of R^* with noise. Optimizing $\hat{R}$ selects for positive noise (optimizer's curse). Effect is $O(\sigma^2/n)$.
2. **Extremal:** $\hat{R}$ and R^* agree over most of policy space but diverge in extreme regions. The optimizer is drawn to precisely these regions.
3. **Causal:** $\hat{R}$ captures a correlate of R^*, not a cause. Optimization breaks the correlation.
4. **Adversarial:** The proxy is deliberately constructed to exploit the optimizer. (Not relevant to our setting.)

Skalse et al. prove: for any $\hat{R} \neq R^*$ and any MDP, there exists a critical optimization threshold β^* such that for $\beta > \beta^*$, the optimized policy has lower true return than the reference.

3.2 Gao et al.'s Scaling Laws (ICML 2023)

The setup. You can't cheaply get human preferences at scale, so Gao et al. use a synthetic setup: a large "gold-standard" reward model R_{gold} plays the role of humans. A smaller "proxy" reward model R_{proxy} is trained on labels from R_{gold}. Then a policy is optimized against R_{proxy} using either reinforcement learning (PPO) or best-of-n sampling. The question: as you optimize harder against the proxy, what happens to the gold score?

The key equation. For both RL and best-of-n, the gold RM score as a function of KL divergence d from the reference policy is well-fit by:

$$R_{\text{gold}}(d) = \alpha\sqrt{d} - \beta \cdot d \tag{4}$$

This has two terms with distinct interpretations:

- $\alpha\sqrt{d}$: the **learning term**. Genuine alignment improvement. Grows as $\sqrt{d}$ — diminishing returns from optimization. This corresponds to the model learning the shared signal between proxy and gold rewards.
- $\beta \cdot d$: the **overoptimization term**. Exploitation of proxy imperfection. Grows *linearly* in d, so it eventually dominates the sublinear learning term.

The gold score peaks at:

$$d^* = \frac{\alpha^2}{4\beta^2} \qquad \text{with peak gold score} \qquad R^*_{\text{gold}} = \frac{\alpha^2}{4\beta} \tag{5}$$

Beyond d^*, every additional unit of optimization *decreases* the true reward. The optimizer is now in Goodhart territory.

Why $\sqrt{d}$ vs. d? The intuition: the beneficial component of the proxy reward corresponds to the *projection* of the proxy onto the true reward direction. In a high-dimensional space, this projection is a lower-dimensional signal that saturates quickly ($\sqrt{d}$). The harmful component—the orthogonal residual—has more degrees of freedom and grows linearly. This is a geometric argument: in D dimensions, the projection onto a 1D truth direction concentrates at rate $1/\sqrt{D}$, while the residual has $D - 1$ dimensions to exploit.

Scaling with reward model size. As the proxy RM gets larger (more parameters), α increases (better proxy $\Rightarrow$ more genuine learning) and β decreases (fewer exploitable flaws). Both effects push d^* rightward — you can optimize harder before hitting Goodhart. Gao et al. find smooth power-law scaling of α and β with RM parameter count.

Scaling with dataset size. More labels for training the proxy RM also push d^* rightward, but with diminishing returns. Doubling the label dataset is less effective than doubling the RM size.

RL vs. best-of-n. Both follow Equation 4, but with different coefficients. The key finding: *RL consumes much more KL than best-of-n for the same amount of optimization.* At the same d_{KL}, best-of-n achieves a higher gold score. RL "wastes" KL budget on distributional changes that don't improve alignment.

KL penalty $\approx$ early stopping. Adding a KL penalty to the RL objective does not change the d-vs-gold-score frontier. Any point on the frontier can be reached by early stopping alone, without the KL penalty. This is a surprising and somewhat discouraging result: the standard RLHF regularization technique doesn't actually buy you anything beyond what early stopping provides.

Connection to our work. Gao et al. parameterize overoptimization by d_{KL} (distance in policy space). We parameterize by N (number of preference pairs). The analogy:

Gao et al.	Our setting
KL divergence d	Training pairs N
Gold RM score	Hidden stakeholder utility
Proxy RM	Misspecified BT weights
Peak at d^*	Sweet spot at N^*
$\alpha\sqrt{d}$: learning	BT convergence toward true signal
$\beta \cdot d$: Goodhart	BT convergence toward misspec. direction

Our $N^* \approx 50$ on MovieLens plays the same role as their d^*: the point where learning saturates and overoptimization begins. A natural question is whether our data admits a similar $\alpha\sqrt{N} - \beta N$ functional form.

3.3 Catastrophic Goodhart (Kwa et al., NeurIPS 2024)

The problem with KL regularization. The standard defense against reward hacking in RLHF is to penalize the KL divergence between the optimized policy and the reference:

$$\max_{\pi} \; \mathbb{E}_{\pi}[\hat{R}(x)] - \lambda \cdot D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}}) \tag{6}$$

The idea: by staying "close" to the reference policy (in KL sense), you can't deviate too far from reasonable behavior, even if $\hat{R}$ is imperfect. Gao et al.'s work implicitly relies on this—their functional form assumes KL controls the damage.

Kwa et al. show this assumption fails under a specific, realistic condition.

The key distinction: light-tailed vs. heavy-tailed error. Define the proxy error $\epsilon(x) = \hat{R}(x) - R^*(x)$. The distribution of ϵ across responses x drawn from the reference policy can be:

- **Light-tailed** (e.g., Gaussian): extreme errors are exponentially rare. $P(|\epsilon| > t) \leq e^{-ct^2}$.
- **Heavy-tailed** (e.g., Pareto, log-normal): extreme errors are polynomially rare. $P(|\epsilon| > t) \sim t^{-\alpha}$.

Theorem (informal): Light tails ⇒ KL works. When ϵ is light-tailed and independent of R^*, optimal policies under decreasing KL penalties achieve arbitrarily high true utility. The KL constraint successfully limits exploitation because extreme proxy rewards (where ϵ is large) are exponentially unlikely, so upweighting them costs exponential KL.

Theorem (informal): Heavy tails ⇒ KL fails catastrophically. When ϵ is heavy-tailed, there exist policies that achieve:

$$\mathbb{E}_{\pi}[\hat{R}] \to \infty \quad \text{while} \quad D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}}) \to 0 \quad \text{and} \quad \mathbb{E}_{\pi}[R^*] \leq \mathbb{E}_{\pi_{\mathrm{ref}}}[R^*] \tag{7}$$

In words: the policy gets arbitrarily high proxy reward with *negligible* KL cost, while achieving *no improvement* in true utility. The KL penalty is useless.

Why does this happen? The key insight is about how KL divergence interacts with tails. To achieve high expected proxy reward, the policy needs to upweight responses where $\hat{R}$ is large. If $\hat{R}$ is large because ϵ is large (not because R^* is large), this is Goodhart.

The KL cost of upweighting a response x by factor t is $\sim \log t$. The proxy reward gain is $\sim \hat{R}(x)$. Under heavy tails, there exist responses with $\hat{R}(x) \sim t^{\alpha}$ but probability $\sim t^{-\alpha}$ under the reference. Upweighting by factor t costs KL $\sim \log t$ but gains reward $\sim t^{\alpha}$. As $t \to \infty$, the reward-to-KL ratio diverges. The optimizer can always find more to exploit, and KL can never constrain it.

Under light tails, the same responses have probability $\sim e^{-ct^2}$, so upweighting by factor t costs KL $\sim ct^2$. The cost grows *faster* than the reward, so eventually KL wins.

Why KL Fails: The Geometry of Average vs. Concentrated

Your intuition is correct: KL is an average measure, and the reward gain is concentrated. Let's make this precise at three levels.

Level 1: The arithmetic. KL divergence is an *expectation* of the log-likelihood ratio:

$$D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}}) = \sum_x \pi(x) \log \frac{\pi(x)}{\pi_{\mathrm{ref}}(x)} \tag{8}$$

Upweighting a response x by factor t (setting $\pi(x) = t \cdot \pi_{\mathrm{ref}}(x)$) contributes:

$$\Delta\mathrm{KL} = t \cdot \pi_{\mathrm{ref}}(x) \cdot \log t \tag{9}$$

The contribution to expected reward is:

$$\Delta\mathrm{Reward} = t \cdot \pi_{\mathrm{ref}}(x) \cdot \hat{R}(x) \tag{10}$$

The reward-to-KL ratio from this single response is $\hat{R}(x)/\log t$. The penalty ($\log t$) is **concave** in the upweighting factor. The reward ($\hat{R}(x)$) can be arbitrarily large. So for any fixed KL budget, you can always find responses where the reward gain exceeds the KL cost—*if* such extreme-reward responses exist with non-negligible probability.

That's the "if" that distinguishes light from heavy tails.

Level 2: Why "average" fails against "concentrated." KL divergence measures the *average* surprise when sampling from π but expecting π_{ref}. It's insensitive to what happens at any *specific* response—it only sees the aggregate.

The optimizer's reward gain is dominated by a *few extreme responses*—the ones where $\hat{R}(x)$ is large. Under heavy tails, these responses have non-negligible probability. The optimizer concentrates its probability budget on them.

The mismatch: KL spreads its penalty across the entire distribution (every response that moved contributes). The reward concentrates its gain on a few extreme points. An average penalty can't constrain a concentrated gain—it's like trying to cap a spike with a blanket.

Under light tails, extreme responses are exponentially rare ($P(\hat{R} > t) \sim e^{-ct^2}$). Upweighting them enough to matter for expected reward requires exponential probability inflation, which costs exponential KL. The blanket is thick enough.

Under heavy tails, extreme responses are polynomially rare ($P(\hat{R} > t) \sim t^{-\alpha}$). Upweighting them costs logarithmic KL but gains polynomial reward. The blanket is too thin.

Level 3: The variational characterization. KL divergence has a precise optimality property (Donsker–Varadhan representation):

$$D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}}) = \sup_f \left\{ \mathbb{E}_\pi[f] - \log \mathbb{E}_{\pi_{\mathrm{ref}}}[e^f] \right\} \tag{11}$$

Rearranging: if $D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}}) \le C$, then for any function f:

$$\mathbb{E}_\pi[f] \le \log \mathbb{E}_{\pi_{\mathrm{ref}}}[e^f] + C \tag{12}$$

Apply this with $f = \hat{R}$ (the proxy reward):

$$\mathbb{E}_\pi[\hat{R}] \le \underbrace{\log \mathbb{E}_{\pi_{\mathrm{ref}}}[e^{\hat{R}}]}_{\text{moment generating function}} + C \tag{13}$$

When $\hat{R}$ is **light-tailed**, the MGF $\mathbb{E}[e^{\hat{R}}]$ is finite, so the bound is meaningful: KL budget C limits the achievable reward.

When $\hat{R}$ is **heavy-tailed**, the MGF is *infinite*: $\mathbb{E}[e^{\hat{R}}] = \infty$. The bound becomes $\mathbb{E}_\pi[\hat{R}] \leq \infty + C = \infty$. It says nothing. KL has zero constraining power.

This is the deepest answer: KL divergence constrains expected values of functions whose **moment generating functions are finite** under the reference distribution. Heavy-tailed rewards violate this condition. No finite KL budget can constrain them.

Would other divergences help?

- **χ^2 divergence:** $\chi^2(\pi\|\pi_{\text{ref}}) = \mathbb{E}_{\pi_{\text{ref}}}[(\pi/\pi_{\text{ref}} - 1)^2]$. Upweighting by t costs $(t-1)^2$—*quadratic*, not logarithmic. Much more effective against concentrated upweighting. But also much more restrictive (rules out many useful policies).
- **Rényi divergence of order $\alpha > 1$:** Interpolates between KL ($\alpha \to 1$) and max-divergence ($\alpha \to \infty$). Higher α penalizes tails more. Using Rényi with $\alpha > 1$ as the regularizer would provide stronger guarantees for heavy-tailed rewards, at the cost of more conservative policies.
- **Total variation:** $\text{TV}(\pi, \pi_{\text{ref}}) = \frac{1}{2}\|\pi - \pi_{\text{ref}}\|_1$. Bounds $|\mathbb{E}_\pi[f] - \mathbb{E}_{\pi_{\text{ref}}}[f]| \leq \|f\|_\infty \cdot \text{TV}$. Works for bounded f but is vacuous for unbounded rewards.

The choice of divergence is a choice of *which deviations you penalize most*. KL penalizes average deviations (good for typical behavior). χ^2 penalizes concentrated deviations (good against tail exploitation). The right choice depends on whether the threat is diffuse or concentrated—which is exactly the light-tail / heavy-tail distinction.

Empirical finding: current reward models appear light-tailed. Kwa et al. test real reward models (Starling 7B, Pythia 1.4B) and find error distributions that are "approximately normal and, therefore, light-tailed." Current RLHF systems are probably safe from catastrophic Goodhart—but future systems with more powerful reward models and optimization might not be.

Connection to our work. Our setting has a specific structure that maps onto this framework:

- Our "proxy" is the misspecified user utility $\hat{\mathbf{w}}_u^\top \phi(x)$.
- Our "true reward" is the diversity utility $\mathbf{w}_d^\top \phi(x)$.
- The "error" is $\epsilon(x) = (\hat{\mathbf{w}}_u - \mathbf{w}_d)^\top \phi(x)$, which is a *linear function* of fixed genre features.
- Since genre features are bounded binary vectors, ϵ is *light-tailed* by construction. Catastrophic Goodhart does not apply.

However, our Goodhart effect is *not* mitigated by this. Why? Because we don't use KL regularization. Our optimizer (BT preference learning) has no KL penalty—it simply converges toward the direction specified by the training labels. The relevant parameter is N (data quantity), not KL divergence. Kwa et al.'s result is about the limits of KL as a *defense*; our result is about the limits of data as an *offense*—more data with wrong specification amplifies the error regardless of distributional tail behavior.

Let's unpack this distinction and its implications.

Defense vs. Offense: Two Different Failure Modes

Kwa et al.'s world (defense failure): There is a fixed, imperfect proxy reward model $\hat{R}$. A policy optimizer (PPO) tries to maximize $\hat{R}$ subject to a KL constraint. The question is whether the constraint (the defense) is strong enough to prevent the optimizer from reaching regions where $\hat{R} \neq R^*$. Under heavy tails: no. Under light tails: yes, but the protected region scales with the KL budget.

The causal chain is: optimizer pushes policy → KL tries to stop it → heavy tails let the optimizer slip through.

Our world (offense failure): There is no policy optimizer. BT preference learning is a *supervised* learner—it fits weights to preference labels via gradient descent. It has no "pressure" to push beyond the training signal; it simply converges toward the direction implied by the labels. The "offense" is the *data itself*: each preference pair is a gradient step toward the misspecified utility direction.

The causal chain is: more data → BT converges more precisely → learned weights align with misspecified direction → content selection changes → hidden stakeholder degrades.

There is nothing to "defend against" because the attack isn't optimization pressure—it's specification error baked into the training signal. KL regularization penalizes how far the policy moves; but in our setting the policy moves *because the data tells it to*, and more data tells it more precisely.

What If We Switched to Softmax Sampling?

Suppose we replaced deterministic top-K selection with probabilistic softmax sampling:

$$P(x \mid \mathbf{w}) = \frac{\exp(\mathbf{w}^\top \phi(x)/\tau)}{\sum_{x'} \exp(\mathbf{w}^\top \phi(x')/\tau)} \tag{14}$$

Now we have a stochastic policy and can define:

$$D_{\mathrm{KL}}(P(\cdot \mid \hat{\mathbf{w}}) \| P(\cdot \mid \mathbf{w}^*)) \tag{15}$$

Could we then use KL to prevent Goodhart?

First observation: We're in the light-tailed regime. Our content features $\phi(x)$ are bounded binary genre vectors, so $\mathbf{w}^\top \phi(x)$ is bounded for any $\mathbf{w}$. The softmax distribution has bounded log-ratios. By Kwa et al.'s Theorem 4 (light tails), KL *would* provide a meaningful constraint. No catastrophic Goodhart.

Second observation: KL would constrain but not eliminate the effect. Adding KL between the learned and reference softmax policies is equivalent to limiting how far the BT-learned weights can stray from some reference weights. But this is just *early stopping* by another name—exactly Gao et al.'s finding that KL penalty ≈ early stopping.

In our N-parameterization, this means: KL regularization would formalize the sweet spot at N^*. It would let you control *where* on the learning-vs-Goodhart curve you operate. But the curve still exists—the tension between user and diversity utility is structural, not an artifact of the optimization method.

Concretely: with softmax + KL, you'd trade the parameter N (data quantity) for the parameter λ (KL penalty strength). At high λ (strong penalty): the policy stays near the reference, learning little—equivalent to low N. At low λ (weak penalty): the policy converges to the misspecified direction—equivalent to high N. The Goodhart curve is reparameterized, not removed.

Third observation: Where would the reference weights come from? In RLHF, the reference is the pretrained model (before fine-tuning). In our setting, the reference would need to be "correct" stakeholder weights—but if we had correct weights, we wouldn't need BT training at all. Using random/initial weights as reference penalizes ALL learning (good and bad), which is wasteful. The fundamental problem is that KL penalizes *distance* from reference, not *direction* of movement. If the movement direction is wrong (misspecified utility), KL can slow you down but can't redirect you.

Heavy Tails, Extremal Goodhart, and Our Setting

The four Goodhart mechanisms and the tail behavior form a two-dimensional picture. Let's connect them:

Regressional Goodhart doesn't care about tails. It's about noise in the proxy estimate, and it gets *better* with more data (the noise averages out). Tail behavior is irrelevant because the effect is at the *center* of the distribution, not the tails.

Extremal Goodhart is about what happens when the optimizer reaches regions where proxy and truth diverge. Two questions:

1. **Do such regions exist?** Yes, whenever proxy $\neq$ truth. In our setting: the region where content is user-optimal but diversity-poor exists because $\cos(\mathbf{w}_u, \mathbf{w}_d) < 0$. These are movies in popular genres with high ratings—they exist in any realistic content pool.

2. **Does the optimizer reach them?** This is where tails matter—but *only for policy optimizers with KL constraints.* Under heavy tails, KL can't stop the optimizer from reaching the extremes. Under light tails, KL bounds how far the optimizer gets.

In our setting, tail behavior is irrelevant to the extremal mechanism. Here's why:

Our "optimizer" is BT preference learning, which converges to the direction specified by the training labels. It doesn't search over a policy space looking for extreme responses. It learns a weight vector that ranks content. The content pool is *finite* (3,347 movies on ML-1M). The "extreme" items—movies that are user-optimal but diversity-poor—are just specific movies in the pool. They're not in any distributional tail; they're fixed items.

As N increases, BT doesn't discover new extremes. It learns to *rank the existing items more accurately* according to the misspecified utility. At $N = 25$, the ranking is noisy and the top-K selection includes a mix of items. At $N = 2000$, the ranking precisely identifies the user-optimal items and puts them at the top.

The mechanism is:

More data $\rightarrow$ more precise ranking $\rightarrow$ top-K concentrates on misspec-optimal items $\rightarrow$ diversity degrades

This is extremal Goodhart—the optimizer reaches the extreme region where proxy and truth diverge. But it reaches it through *convergence of a supervised learner*, not through *exploitation of distributional tails*. The finite content pool removes any role for tail behavior.

To summarize the relationship:

	RLHF setting	**Our setting**
What causes Goodhart?	Policy optimizer finds extreme responses where $\hat{R} \neq R^*$	BT converges to misspecified direction; top-K concentrates on wrong content
Mechanism	Extremal: optimizer reaches tail of distribution	Extremal: supervised learner precisely learns wrong ranking over finite pool
Role of tail behavior	Heavy tails $\Rightarrow$ KL can't prevent reaching extremes	Irrelevant: content pool is finite, no distributional tails
What controls severity?	KL budget / tail heaviness	N (data quantity) / $\cos(\mathbf{w}_{\text{target}}, \mathbf{w}_{\text{hidden}})$
Defense	KL penalty (limited by tails)	Early stopping on N / independent selection dim (δ)

Both settings exhibit extremal Goodhart. The *channel* through which the optimizer reaches the extreme region is different: policy optimization in RLHF, supervised convergence in BT preference learning. The tail behavior of the reward distribution governs the first channel but is irrelevant to the second. In both cases, the core issue is the same: the optimizer finds exactly what it's told to find, and if what it's told is wrong, precision makes things worse.

3.4 What None of These Address

- **Multiple objectives with known geometry.** All above treat one proxy vs. one true reward. None study settings where multiple stakeholders have utility vectors with known cosine similarity.
- **Data-dependent thresholds.** Gao et al. parameterize by KL divergence, not by the number of preference pairs N. The sweet spot in N (our finding: $N \approx 50$ on MovieLens) is not studied.
- **Selection mechanism independence.** Our rank stability finding (= 1.0 for individual parameter perturbation) initially suggested the Pareto frontier "buffers" against Goodhart. The actual mechanism is more specific—see section 3.6 below.
- **Low-rank weight space.** Our Phase 9 finding that the weight space is effectively 3-dimensional (projection cosine > 0.97) constrains the Goodhart dynamics—the misspecification can only push within a low-dimensional subspace.

3.5 Why KL Divergence Doesn't Appear in Our Formulation

Every paper above parameterizes overoptimization by KL divergence between policies. We parameterize by N (preference pairs). This isn't an oversight—it reflects a structural difference in our optimization pipeline.

In RLHF: The policy $\pi(y \mid x)$ is a distribution over responses. KL divergence $D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}})$ measures how far the optimized distribution has moved from the reference. The optimizer (PPO) is explicitly regularized by KL.

In our setting: The "policy" is a deterministic function: given weight vector $\mathbf{w}$, select top-K content by $\mathbf{w}^\top \phi(x)$. There is no stochastic policy, no reference distribution, and no KL penalty. The BT optimizer simply minimizes preference loss on N pairs until convergence.

What plays the role of KL? In our setting, the **cosine distance** between the learned and true weight vectors:

$$d_{\cos} = 1 - \cos(\hat{\mathbf{w}}(N), \mathbf{w}^*) \tag{16}$$

serves the same conceptual role as d_{KL}. Both measure "how far the optimizer has moved from the reference." The difference:

- d_{KL} measures distance in *policy distribution* space (infinite-dimensional, information-geometric).
- $d_{\cos}$ measures distance in *weight direction* space (D-dimensional, Euclidean geometry on the unit sphere).

Why this matters for Goodhart. In RLHF, KL controls optimization pressure—larger KL budget ⇒ more optimization ⇒ more Goodhart. In our setting, N controls optimization pressure—more pairs ⇒ BT converges more precisely ⇒ $d_{\cos}$ between $\hat{\mathbf{w}}(N)$ and the *misspecified* target increases (the model moves further from the true direction toward the wrong direction).

The Gao et al. functional form $R_{\mathrm{gold}}(d) = \alpha\sqrt{d} - \beta d$ could potentially be re-expressed in our setting as $U_{\mathrm{hidden}}(d_{\cos}(N))$, where $d_{\cos}(N)$ captures the BT convergence trajectory. The intermediate variable $d_{\cos}$ would bridge between N (our control parameter) and U_{hidden} (our outcome), with the $\sqrt{\cdot}$ term capturing the learning phase and the linear term capturing the Goodhart phase.

Could we add KL to our setting? Yes—by replacing hard top-K selection with softmax sampling:

$$P(x \mid \mathbf{w}) = \frac{\exp(\mathbf{w}^\top \phi(x)/\tau)}{\sum_{x'} \exp(\mathbf{w}^\top \phi(x')/\tau)} \tag{17}$$

Then $D_{\mathrm{KL}}(P(\cdot \mid \hat{\mathbf{w}}) \| P(\cdot \mid \mathbf{w}^*))$ is well-defined and connects directly to Gao/Kwa's framework. For small weight perturbations, this KL relates to cosine distance via the Fisher information of the softmax. This is a

potential direction for formalization—but it changes the content selection mechanism from deterministic to stochastic, which may affect the Goodhart dynamics.

3.6 The "Pareto Buffer": What It Actually Is

Our rank stability = 1.0 finding initially suggested the Pareto frontier "buffers" against Goodhart. On closer examination, the mechanism is more specific and less about Pareto geometry than about **selection mechanism independence**.

Two ways to select an operating point:

1. **Selection via misspecified weights (Goodhart possible).** Use learned weights as the content scorer. The misspecification directly determines what gets recommended. More precise misspecification $\Rightarrow$ more precisely wrong selection $\Rightarrow$ hidden stakeholder degrades. This is our Phase 6 Goodhart experiment.

2. **Selection via independent parameter (buffer exists).** Pick the operating point using the diversity weight δ, which is *independent* of any stakeholder's utility weights. Perturbing weights shifts the utility *values* at each δ but doesn't change *which* δ you choose. This is the rank stability experiment.

Why rank stability = 1.0: The diversity weight δ controls the tradeoff between engagement and topic diversity. It's an independent control knob—not derived from any stakeholder utility. When you perturb a single utility weight, the utility *values* at each δ shift, but the dominance relations among δ-points don't change. The same 7 of 21 δ-points remain Pareto-optimal.

When the buffer fails:

- **Correlated perturbation:** Simultaneously perturbing multiple weights changes utility values by different amounts at different δ-points, breaking the dominance ordering. Rank stability drops below 1.0.

- **Misspecified scorer:** If the learned (misspecified) weights are used as the content *scorer* (not just the evaluator), the selection itself changes. δ still controls the diversity-engagement tradeoff, but the engagement component now points in the wrong direction.

Connection to Gao et al.'s KL $\approx$ early stopping result: KL regularization in RLHF is supposed to be an independent control (constraining how far the policy moves), but Gao et al. found it's redundant with the optimization axis—KL penalty doesn't change the KL-vs-gold-score frontier. Our diversity weight δ succeeds where KL fails because δ is *genuinely orthogonal* to the stakeholder utility weights. It controls content *diversity* (a structural property of the recommendation set), not content *quality* (which depends on the utility direction). The Pareto structure helps not because it's "Pareto" but because it provides a control dimension that the optimizer can't corrupt.

The real insight: Separating the selection mechanism (diversity weight) from the specification mechanism (stakeholder utility weights) provides protection against individual misspecification. This is a design principle, not a mathematical theorem about Pareto frontiers. It suggests that recommendation systems should maintain at least one control dimension (like diversity) that is defined structurally (e.g., set-level entropy) rather than derived from stakeholder utility weights.

4 Our Empirical Results

4.1 The Multi-Stakeholder Goodhart Curve

Three stakeholders: user ($\mathbf{w}_u$), platform ($\mathbf{w}_p$), diversity ($\mathbf{w}_d$), with:

$$\cos(\mathbf{w}_u, \mathbf{w}_d) \approx -0.3, \quad \cos(\mathbf{w}_u, \mathbf{w}_p) \approx 0.95 \tag{18}$$

We perturb user weights: $\hat{\mathbf{w}}_u = \mathbf{w}_u \odot (1 + \mathcal{N}(0, \sigma))$ with $\sigma = 0.3$, train BT on N preference pairs labeled by $\hat{\mathbf{w}}_u$, and measure true diversity utility of the selected content.

N	User utility (target)	Diversity utility (hidden)
25	18.68	−6.70
50	19.44	**−6.34** *(peak)*
100	20.08	−7.24
200	20.81	−7.63
500	21.24	−8.38
2000	21.33	−8.98

User utility increases monotonically (+14%). Diversity utility peaks at $N = 50$, then degrades 42% by $N = 2000$. This is textbook **extremal Goodhart**: the misspecified utility and the hidden utility are anti-correlated, so more precise optimization of one systematically harms the other.

4.2 Key Empirical Findings

1. **The sweet spot exists and is data-dependent.** $N \approx 50$ on MovieLens (genre features, 19-dim). $N \approx 100$ on synthetic (action features, 18-dim). The sweet spot is where BT has learned the overlapping signal but hasn't yet converged to the misspecified direction.

2. **The mechanism is utility transfer, not noise.** As N increases, the content selection shifts from "approximately random" to "precisely user-optimal," which is anti-optimal for diversity ($\cos < 0$). This is systematic, not regressional.

3. **Strategy 1 (better specification) confirms the mechanism.** Even with $\sigma = 0$ (correct specification), the user-optimal scorer selects content that harms diversity. Goodhart amplifies a tension that exists even under correct specification.

4. **The weight space is low-rank.** Phase 9 showed projection cosine > 0.97 for scalarized $\rightarrow$ per-stakeholder span. The Goodhart dynamics are confined to a 3D subspace of the 19D genre feature space.

5 Literature Map

Question	Status	Reference
Does Goodhart occur in RL?	Answered (yes, provably)	Skalse et al. 2024
Scaling laws for overoptimization?	Answered (KL-based)	Gao et al. 2023
Does KL regularization help?	Answered (no, heavy tails)	Kwa et al. 2024
BT preference consistency?	Answered (order-consistent)	Sun et al. 2025
Multi-stakeholder Pareto?	Framework exists	Burke & Abdollahpouri 2017
Multi-stakeholder Goodhart?	**Open — our work**	
N as Goodhart modulator?	**Open — our work**	
Stakeholder opposition (cos < 0)?	**Open — our work**	
Pareto as Goodhart buffer?	**Open — our work**	
Low-rank weight constraints?	**Open — our work**	

6 Toward Formalization

6.1 Setup

Let K stakeholders have utility directions $\mathbf{w}_1, \ldots, \mathbf{w}_K \in \mathbb{R}^D$. A content pool $\{\phi(x)\}_{x=1}^M$ with feature vectors $\phi(x) \in \mathbb{R}^D$. A BT model learns $\hat{\mathbf{w}}(N)$ from N preference pairs labeled by a (possibly misspecified) utility.

Content selection: $S(\mathbf{w})$ = top-k by $\mathbf{w}^\top \phi(x)$.

Hidden stakeholder utility: $U_{\text{hidden}}(S) = \frac{1}{k} \sum_{x \in S} \mathbf{w}_{\text{hidden}}^\top \phi(x)$.

6.2 The Goodhart Condition

Proposition 1 (Informal)**.** *More data hurts the hidden stakeholder when the learned direction moves* away *from the hidden utility direction:*

$$\frac{\partial U_{hidden}(S(\hat{\mathbf{w}}(N)))}{\partial N} < 0 \quad \Longleftrightarrow \quad \nabla_{\mathbf{w}} U_{hidden} \cdot \frac{\partial \hat{\mathbf{w}}}{\partial N} < 0 \tag{19}$$

The left side is the chain rule: how hidden utility changes with N, mediated by how the learned weights change the selection and how the selection affects hidden utility.

Unpacking the two terms:

- $\nabla_{\mathbf{w}} U_{\text{hidden}}$: How does hidden utility change as the scorer direction shifts? This is approximately $\mathbf{w}_{\text{hidden}}$ projected onto the content feature space — it points toward content that's good for the hidden stakeholder.
- $\frac{\partial \hat{\mathbf{w}}}{\partial N}$: The direction BT converges toward as N increases. With misspecified labels, this points toward $\hat{\mathbf{w}}_{\text{misspec}}$ — the misspecified utility direction.

The Goodhart condition:

$$\cos(\mathbf{w}_{\text{hidden}}, \hat{\mathbf{w}}_{\text{misspec}}) < 0 \tag{20}$$

More data hurts when the hidden stakeholder's utility direction is anti-correlated with the direction BT is converging toward. In our setting: $\cos(\mathbf{w}_{\text{diversity}}, \hat{\mathbf{w}}_{\text{user}}) \approx -0.3 < 0$. Goodhart is predicted, and confirmed empirically.

6.3 Numerical Validation (Phase 10)

The direction condition is validated on 4 data points across 2 datasets, with zero violations:

Dataset	Hidden	$\cos(\mathbf{w}_u, \mathbf{w}_h)$	Predicted	Actual
ML-100K	platform	+0.956	Improve	+14% ✓
ML-100K	diversity	−0.313	Degrade	−42% ✓
ML-1M	platform	+0.926	Improve	+14% ✓
ML-1M	diversity	−0.155	Degrade	−46% ✓

Direction condition holds perfectly. Every stakeholder with $\cos < 0$ degrades with N; every stakeholder with $\cos > 0$ improves. Platform ($\cos \approx +0.95$) serves as the positive control: it improves monotonically because BT converges toward a direction *aligned* with platform utility.

Magnitude condition is more nuanced. ML-1M shows more degradation (46%) than ML-100K (42%) despite *less* anti-correlation (−0.155 vs −0.313). The likely explanation: ML-1M has more content (3,347 vs 1,305 movies), giving the optimizer more user-optimal/diversity-poor items to concentrate on. Degradation severity depends on both the angle between utility directions *and* the geometry of the content pool—the angle determines the *direction* of degradation, and the content pool determines how much utility can be transferred in that direction.

Statement of the validated condition:

Proposition 2 (Multi-Stakeholder Goodhart Direction Condition)**.** *In a linear BT preference learning setting with K stakeholders and a finite content pool, let $\hat{\mathbf{w}}_t$ be the (possibly misspecified) target utility learned from N preference pairs, and $\mathbf{w}_h$ be a hidden stakeholder's true utility. Then:*

- *If $\cos(\hat{\mathbf{w}}_t, \mathbf{w}_h) < 0$: $U_h(S(\hat{\mathbf{w}}_t(N)))$ is eventually decreasing in N.*
- *If $\cos(\hat{\mathbf{w}}_t, \mathbf{w}_h) > 0$: $U_h(S(\hat{\mathbf{w}}_t(N)))$ is eventually increasing in N.*

Validated on 3 datasets (synthetic, ML-100K, ML-1M) with 0 violations.

6.4 Open Questions for Formalization

1. **Rate of degradation.** How fast does U_{hidden} decline with N? Is it $O(1/\sqrt{N})$ (BT convergence rate) or slower/faster?

2. **Sweet spot location.** Can we predict N^* (the peak of U_{hidden}) from the cosine similarity and feature dimensionality? Our data: $N^* \approx 50$ on ML-100K (19-dim), $N^* \approx 25$ on ML-1M (19-dim, more content).

3. **Pareto buffer.** Why does rank stability = 1.0 for individual parameter perturbation but < 1.0 for correlated perturbation? Can we characterize the perturbation directions that the Pareto frontier absorbs vs. amplifies?

4. **Low-rank constraint.** With K stakeholders in D dimensions, the effective weight space is K-dimensional (our Phase 9 finding). Does this bound the Goodhart severity? Intuitively, a 3D Goodhart in 19D is less severe than a 19D Goodhart — fewer directions to exploit.

5. **The "fix spec first" prescription.** Our data shows: data helps when specification is correct, data hurts when wrong. Can we formalize the *crossover*: at what σ (misspecification level) does additional data switch from beneficial to harmful?

7 References

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8 Appendix: Unpacking Skalse et al.'s Four Mechanisms

Skalse et al. (ICLR 2024) identify four ways that optimizing a proxy reward $\hat{R}$ can decrease the true reward R^*. These aren't competing theories—they're four *mechanisms* that can co-occur. Understanding each one requires thinking about WHY the proxy and reality diverge, not just THAT they diverge.

8.1 Regressional Goodhart

The everyday version. You're hiring and use SAT scores as a proxy for job performance. Among 1000 applicants, the person with the highest SAT score is more likely to have gotten lucky on test day than to be genuinely the smartest. By selecting the extreme, you're selecting for noise.

The math. Suppose the proxy is a noisy measurement of truth:

$$\hat{R}(\pi) = R^*(\pi) + \epsilon(\pi), \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{21}$$

where ϵ is independent noise that varies across policies. The optimizer selects:

$$\pi^* = \arg\max_{\pi} \left[R^*(\pi) + \epsilon(\pi)\right] \tag{22}$$

This selects for policies where both R^* and ϵ happen to be large. The expected true reward of the selected policy is:

$$\mathbb{E}[R^*(\pi^*)] = \max_{\pi} R^*(\pi) - \underbrace{\mathbb{E}\left[\epsilon(\pi^*) \mid \pi^* = \arg\max(\hat{R})\right]}_{\text{optimizer's curse}>0} \tag{23}$$

The stronger the optimization (searching over more policies, or optimizing more aggressively), the more you select for noise. This is exactly the **winner's curse** from auction theory.

Key properties:

- The proxy is *unbiased*—$\mathbb{E}[\hat{R}] = R^*$ for any fixed policy. The problem is only in the *selection*.
- The effect scales as $O(\sigma^2/n)$ where n is the number of samples used to estimate $\hat{R}$. More data reduces it.
- This is the *mildest* form of Goodhart. It vanishes with perfect measurement.

In our setting: Regressional Goodhart would occur if BT training noise caused us to select "lucky" content that happened to score high due to noise in the learned weights. With 18–19 features and 2000+ preference pairs, this effect is small (our held-out accuracy is within 2% of training accuracy). It's not the dominant mechanism.

8.2 Extremal Goodhart

The everyday version. A doctor uses blood pressure as a proxy for heart health. For most patients, lower blood pressure = healthier heart. But at the *extreme*—a patient with blood pressure of 60/40—the correlation breaks. The proxy and reality agreed in the normal range but diverge at the tail.

The math. The proxy and truth are correlated but not identical:

$$\hat{R}(\pi) = f(R^*(\pi)) + g(\pi) \tag{24}$$

where f is monotonically increasing (the correlation) and g captures features of π that affect the proxy but not the truth. In the normal range of π, f dominates. At the extremes, g can dominate.

The optimizer pushes π into extreme regions where $g(\pi)$ is large. The proxy says this is a great policy (high $\hat{R}$). The truth says otherwise.

Remark 1. *This differs from regressional Goodhart in a crucial way: the divergence is* systematic*, not random. The proxy is biased in extreme regions, not just noisy. More data doesn't fix it—the bias is structural.*

Key properties:

- Requires the optimizer to reach extreme regions of policy space. Weak optimization may never trigger it.
- The divergence is *predictable* if you know how g behaves at extremes.
- More data makes it *worse* (the optimizer more precisely finds the extreme region). This is the "U-shaped curve" in true performance.

In our setting: This is the **dominant mechanism**. The "proxy" is the misspecified user utility $\hat{\mathbf{w}}_u$. The "truth" is the diversity utility $\mathbf{w}_d$. They're correlated (both are functions of genre features) but anti-correlated at the extremes ($\cos \approx -0.3$). As BT learns $\hat{\mathbf{w}}_u$ more precisely (more data), it pushes content selection into regions that are *great* for the misspecified user but *terrible* for diversity. The divergence is systematic: it's driven by the angle between utility directions, not by noise.

Why this is our mechanism:

1. The effect gets *worse* with more data (42% degradation at $N = 2000$ vs. peak at $N = 50$). Regressional Goodhart would get *better* with more data.
2. The direction of degradation is predictable from the cosine similarity: diversity degrades because $\cos(\mathbf{w}_d, \hat{\mathbf{w}}_u) < 0$.
3. Strategy 1 (better specification, $\sigma \to 0$) *also* harms diversity, confirming the effect is about the inherent opposition between stakeholders, not about specification noise.

8.3 Causal Goodhart

The everyday version. A hospital measures recovery rates as a proxy for care quality. Hospital A starts refusing to admit severely ill patients. Recovery rates soar. Care quality doesn't change—the hospital just gamed the input distribution.

More precisely: recovery rate *correlates* with care quality in the observational data. But the causal structure is:

$$\text{care quality} \to \text{recovery}$$
$$\text{patient severity} \to \text{recovery}$$

Optimizing recovery by manipulating patient severity breaks the correlation without changing care quality. The proxy captured an association, not a causal relationship.

The math. Let R^* be the true reward and X a confounding variable:

$$\hat{R}(\pi) = h(R^*(\pi), X(\pi)) \tag{25}$$

In observational data, $\hat{R}$ and R^* are correlated because X covaries with both. But the optimizer can change X independently of R^*, breaking the correlation.

Formally, the problem is that $\arg\max_\pi \hat{R}(\pi) \neq \arg\max_\pi R^*(\pi)$ because the optimizer exploits the X pathway rather than the R^* pathway.

Key properties:

- Requires an *actionable* confounder—a variable the optimizer can manipulate that affects the proxy but not the truth.
- Interventional data (randomized experiments) break the confounding. Observational proxies are vulnerable; causal proxies are not.
- This is arguably the most common form of Goodhart in real organizations (gaming metrics by manipulating inputs).

In our setting: Causal Goodhart is *not* the primary mechanism. Our BT model operates on genre features, which are fixed properties of movies—the optimizer can't manipulate them. There's no confounder being exploited. The model selects content based on fixed features, not by gaming an input distribution.

However, in a *production* recommendation system, causal Goodhart is very real: creators gaming the algorithm (e.g., adding trending keywords to boost visibility without improving content quality). Our synthetic benchmark doesn't capture this because the content pool is fixed.

8.4 Adversarial Goodhart

The everyday version. A teacher designs a test (proxy for knowledge). A student obtains the answer key and memorizes it. The student scores perfectly on the test while learning nothing. The proxy was designed in good faith, but an adversary deliberately exploited it.

The math. An adversary $\mathcal{A}$ observes the proxy $\hat{R}$ and constructs a policy $\pi_{\mathcal{A}}$ that maximizes $\hat{R}$ while minimizing R^*:

$$\pi_{\mathcal{A}} = \arg\max_{\pi} \left[\hat{R}(\pi) - \lambda R^*(\pi)\right] \tag{26}$$

This is the worst case: the misalignment is intentional and targeted.

Key properties:

- Requires an intelligent adversary, not just a misaligned proxy.
- The severity is unbounded—the adversary can always find the worst-case exploitation.
- Defenses require robustness guarantees, not just better proxies.

In our setting: Not applicable. Our stakeholders are passive utility functions, not strategic agents. No one is deliberately trying to exploit the reward model. (Though in production platforms, adversarial Goodhart from bot networks and engagement farms is very real.)

8.5 Summary: Which Mechanism Drives Our Results?

Mechanism	Data helps?	In our setting?	Evidence
Regressional	Yes ($\downarrow$ with N)	Minor	Held-out gap < 2%
Extremal	**No ($\uparrow$ with N)**	**Dominant**	**42% degradation,** cos < 0
Causal	Depends	Not in benchmark	Fixed content pool
Adversarial	N/A	No	No strategic agents

The multi-stakeholder setting makes extremal Goodhart especially sharp: the "extreme region" where the proxy diverges from truth isn't some obscure corner of policy space. It's the *entire region where content is user-optimal but diversity-poor*—and BT is designed to find exactly that region as N increases.

This is what makes our contribution distinct from Skalse et al.: in their single-objective framework, you need the proxy to be imperfect in a specific way for extremal Goodhart to occur. In our multi-stakeholder framework, the "imperfection" is *structural*—it's the angle between stakeholder utilities. As long as $\cos(\mathbf{w}_{\text{target}}, \mathbf{w}_{\text{hidden}}) < 0$, more data *always* hurts the hidden stakeholder. The Goodhart effect is not a pathology—it's a geometric consequence of opposing utilities.